

PRODUCT REQUIREMENTS DOCUMENT

Transaction Anomaly Alert System

AI-Powered Anti-Money Laundering (AML) Transaction Screening

Author: Vamsi Krishna Gali | MSc Computing Science, Griffith College Dublin | 2026

1. Document Overview

This Product Requirements Document (PRD) defines the purpose, scope, requirements, and success criteria for the Transaction Anomaly Alert System — an AI-powered Anti-Money Laundering (AML) transaction screening solution. The system is designed to support compliance teams within financial institutions by automatically classifying transaction risk and surfacing high-risk activity for human review.

2. Problem Statement

Financial institutions in Ireland operate within one of Europe's largest financial hubs, exposing them to significant money-laundering risk through cross-border transaction flows.

Traditional manual transaction monitoring is reactive and inefficient — suspicious activity is often identified only after fraud has occurred, forcing investigations to begin from scratch with limited context.

There is a need for an automated, intelligent screening layer that proactively evaluates transactions against key risk indicators — including account age, transaction amount, and transaction frequency — and flags high-risk activity in real time. By identifying patterns such as newly opened accounts processing unusually large transactions, the system enables compliance teams to prioritise genuine threats and initiate investigations earlier in the fraud lifecycle.

3. Objectives and Success Metrics

Primary Objective

To provide compliance teams with an automated, real-time risk classification layer that flags high-risk transactions immediately upon execution, enabling faster and more targeted investigations.

Success Metrics

- **Detection Speed:** Risk classification delivered instantly upon transaction processing, eliminating manual review delays.
- **Risk Accuracy:** Consistent, explainable risk classifications aligned with defined AML threshold criteria, with each flag supported by clear reasoning.
- **Operational Efficiency:** Significant reduction in manual transaction monitoring effort, allowing analysts to focus on genuine high-risk cases.
- **Explainability:** Every flagged transaction includes a transparent rationale, supporting audit and regulatory requirements under the EU AI Act.

4. Users and Personas

Primary User: AML Analyst

The AML Analyst is responsible for monitoring transactions, investigating suspicious activity, and ensuring regulatory compliance within the financial institution. The analyst reviews transactions flagged as high-risk by the AI screening layer. Each flag includes a risk classification and supporting rationale, enabling the analyst to prioritise which accounts require deeper investigation.

Analyst Workflow

- The system flags a transaction as high-risk.

- The AML Analyst reviews the flag and accompanying reasoning.
- The analyst conducts a detailed investigation of the flagged account.
- Findings are escalated to Senior Compliance for further action or regulatory reporting.

Secondary User: Senior Compliance Officer

Receives escalated cases from AML Analysts, makes final determinations on suspicious activity, and oversees regulatory reporting obligations such as Suspicious Activity Reports (SARs).

5. Requirements

Functional Requirements

FR1 — Data Ingestion: The system shall read transaction records including transaction ID, amount, country, account age, and transaction frequency.

FR2 — Risk Evaluation: The system shall evaluate each transaction against defined AML threshold criteria, including amount limits, account age, and transaction frequency patterns.

FR3 — Risk Classification: The system shall classify each transaction as Low, Moderate, or High risk and flag high-risk transactions for analyst review.

FR4 — Result Output: The system shall record the risk classification and supporting rationale against each transaction for analyst access.

Non-Functional Requirements

NFR1 — Performance: The system shall evaluate transactions against threshold criteria with minimal latency, supporting timely review.

NFR2 — Explainability: The system shall provide a clear, human-readable justification for each risk classification, supporting audit and regulatory transparency.

NFR3 — Data Privacy: The system shall process transaction data in a manner that protects sensitive financial information, in alignment with GDPR and EU AI Act requirements.

6. Risks and EU AI Act Compliance

Identified Risks

R1 — False Positives: A legitimate customer may open a new account and deposit substantial genuine savings as their first transaction. The system may incorrectly flag this as high-risk, triggering unnecessary investigation and potentially causing customer dissatisfaction.

R2 — Limited Contextual Understanding: The system classifies risk based on numerical threshold comparisons rather than the underlying purpose of a transaction, which may result in false flags or incorrect classifications.

R3 — Potential Bias: The model may exhibit bias toward certain patterns — such as treating all new accounts or specific transaction types as inherently higher-risk — leading to unfair outcomes.

R4 — Over-Reliance on Automation: Bank staff hold contextual information the system lacks. Excessive reliance on automated flags without human context could lead to poor decisions.

EU AI Act Compliance Considerations

C1 — High-Risk Classification: AML transaction screening processes sensitive financial data and influences consequential decisions, placing it within the high-risk AI category under the EU AI Act. This mandates strict governance, documentation, and oversight.

C2 — Transparency: Customers have a right to be informed that their transactions are subject to automated monitoring, in line with the EU AI Act's transparency obligations.

C3 — Human Oversight: The system must operate as a decision-support tool, not an autonomous decision-maker. All high-risk flags require human review by a qualified AML Analyst before action.

7. Success Criteria

SC1 — Scalable Transaction Monitoring: The system effectively screens high transaction volumes that would be impractical for analysts to review manually, reducing dependence on traditional manual monitoring.

SC2 — Operational Time Savings: The institution realises measurable time and resource savings by automating the initial risk-screening stage of the AML workflow.

SC3 — Focused Analyst Effort: AML Analysts concentrate their expertise on reviewing high-risk flagged transactions rather than examining every transaction, improving both efficiency and effectiveness.

SC4 — Improved Investigation Prioritisation: The institution can identify and prioritise genuinely suspicious activity earlier in the fraud lifecycle, supporting faster and more targeted investigations.

8. Technical Implementation

The prototype was implemented using n8n as the no-code workflow orchestration platform, with the Groq large language model (llama-3.3-70b-versatile) providing risk analysis. Transaction data is managed in Google Sheets. The workflow follows a sequential pipeline: Manual Trigger, Data Ingestion from Google Sheets, AI Risk Analysis, and Result Output written back to the data source with separate Risk Level and Reasoning fields.

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